Admission-Time Prediction of Myocardial Infarction Relapse

Data Analytics Project Report

September 5, 2026

Abstract

Predicting relapse at hospital admission for myocardial infarction would allow earlier preventive care, but admission-time information is limited. On 1700 UCI records (159 relapses, 9.35%) we compare three course-approved classifiers under a leakage-safe protocol: preprocessing is fitted fold-locally, the threshold is locked on validation, and a single linear SVC is evaluated once on held-out test data. It detects 8 of 32 relapses at the cost of 62 false alarms (balanced accuracy `0.5244`, ROC-AUC `0.5597`). The contribution is methodological: an honest, fully auditable baseline showing how weak the admission-time signal is, rather than an inflated score.

1 Dataset description

The dataset contains `1700` hospital records and `124` columns [1]. The analysis predicts relapse of myocardial infarction (`REC_IM`, column `122`) from information available at admission. The target contains `1541` non-relapse and `159` relapse observations, or `90.65%` and `9.35%`, respectively.

The admission-time feature window follows the dataset specification: input columns 2–112 are eligible except columns `93`, `94`, `95`, and 100–105. The identifier and all eleven non-target outcome columns are also excluded. The resulting raw candidate matrix contains `102` predictors. The complete decision record appears in `report/tables/admission_exclusion_audit.csv`.

2 Data cleaning

The raw "?" markers are loaded as missing values. Before any learned cleaning, the admission-safe matrix is split into `1360` development and `340` test rows with stratification and `random_state=42`. The development partition is then split into `1020` fit and `340` validation rows.

Within cross-validation, every model/parameter/fold evaluation learns missingness selection, imputation, clipping, and scaling only from its subfold training rows. The subfold validation rows are transformed without recalculation. After CV selection, one final preprocessor is fitted on the `1020` fit rows and reused unchanged for validation and test. It removes five predictors above the `40%` missingness threshold: `KFK_BLOOD` (`99.90%`), `IBS_NASL` (`96.27%`), `D_AD_KBRIG` and `S_AD_KBRIG` (both `63.53%`), and `NOT_NA_KB` (`40.39%`). Low-cardinality predictors receive fit-derived mode imputation; other predictors receive fit-derived median imputation. Fit, validation, and test therefore share the same `97` predictors and contain no missing values after transformation.

Continuous predictors are clipped to fit-derived $[\mu - 3\sigma, \mu + 3\sigma]$ bounds without dropping observations [3]. The learned fill values and clipping bounds are recorded in `report/tables/feature_handling.csv` and `report/tables/sigma_clipping_summary.csv`. No missingness rule, imputation value, clipping bound, or scaling parameter is fitted on validation or test data.

3 Exploratory analysis

All target-linked exploratory analysis is restricted to fit data so that validation and test labels cannot influence analytical choices. Figure 1 illustrates why plain accuracy is not an adequate primary criterion. Figure 2 reports missingness in the same fit partition.

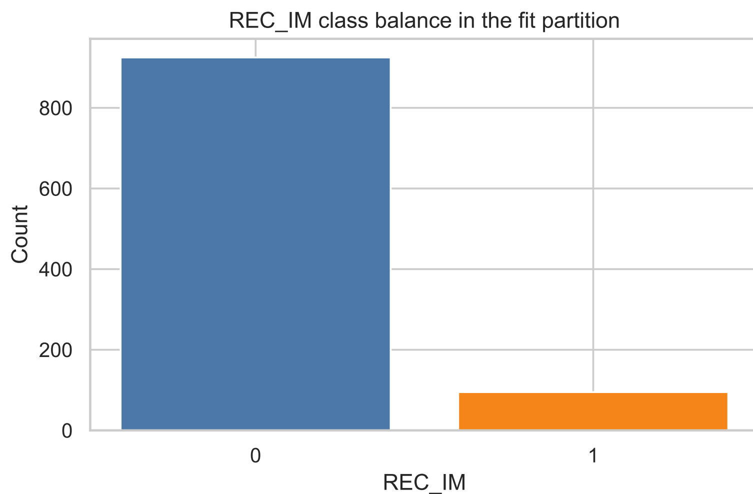

Figure 1: Fit-partition distribution of `REC_IM`: 925 non-relapse and 95 relapse cases.

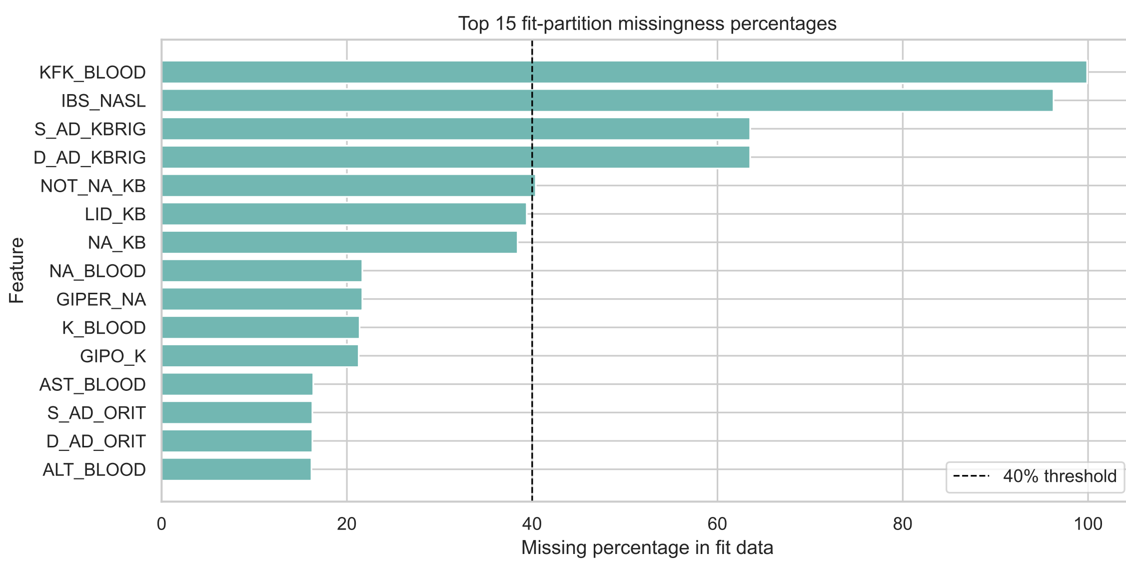

Figure 2: Top-15 missingness percentages in the fit partition. The dashed line marks the 40% removal threshold; five predictors exceed it.

The strongest simple correlation with the target is `zab_leg_02` (`0.1184`), followed by `AGE` (`0.1057`), `STENOK_AN` (`0.0979`), and `np_01` (`0.0978`). Their small magnitudes suggest weak marginal signal. The complete values are exported in `report/tables/top_correlations.csv`; Figures 3 and 4 provide the corresponding descriptive views.

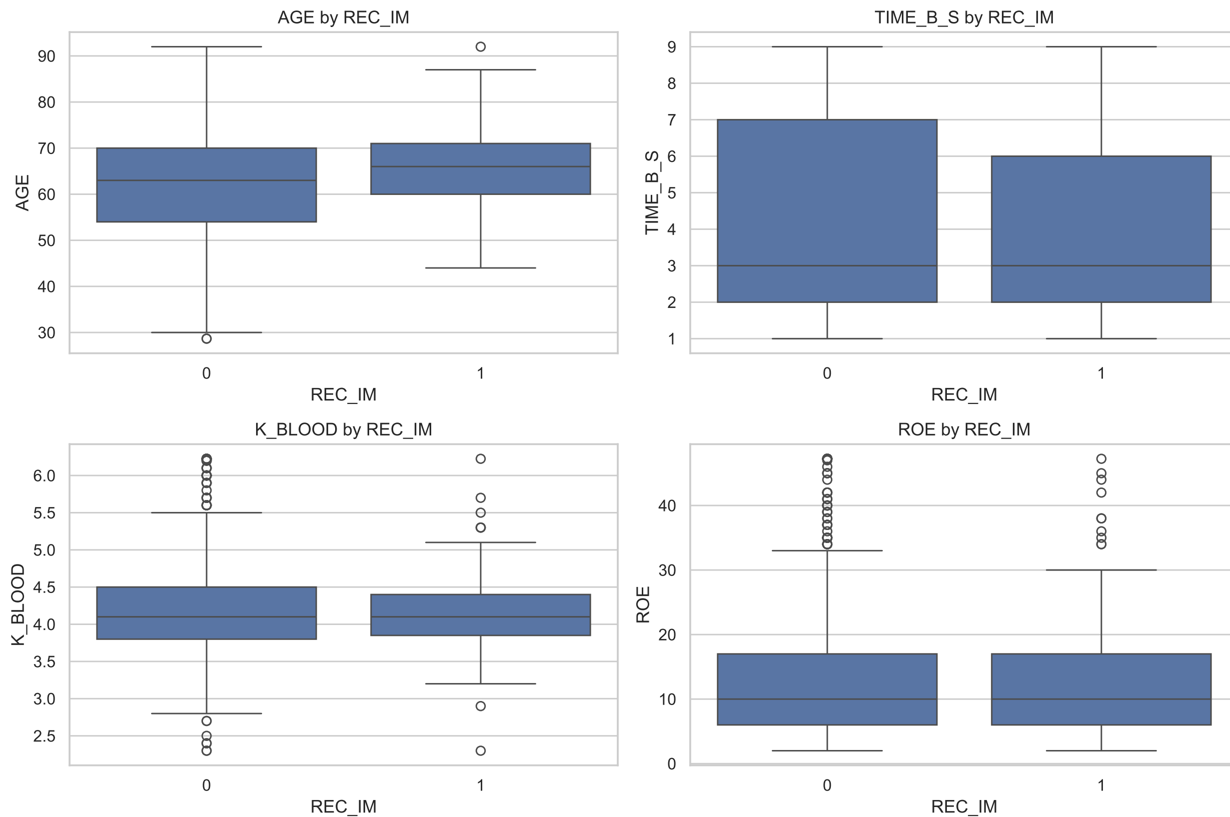

Figure 3: Selected admission-time predictor distributions by `REC_IM` class.

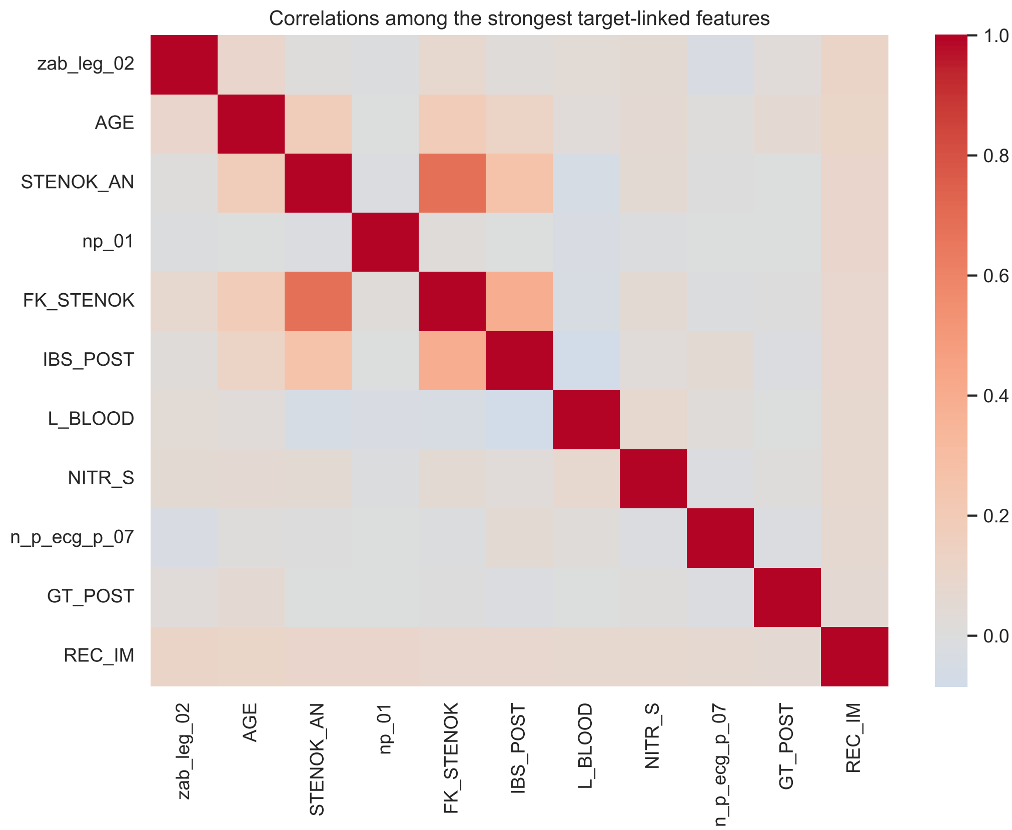

Figure 4: Correlation matrix for the strongest target-linked predictors and `REC_IM`.

3.1 Unsupervised exploration

K-means is fitted to the scaled descriptive feature matrix for exploratory phenotyping. The silhouette criterion selects $K = 2$ with score `0.2599`. The fit-partition clusters contain `104` and `916` observations, with relapse rates `5.8%` and `9.7%`. Figure 5 reports the full sweep.

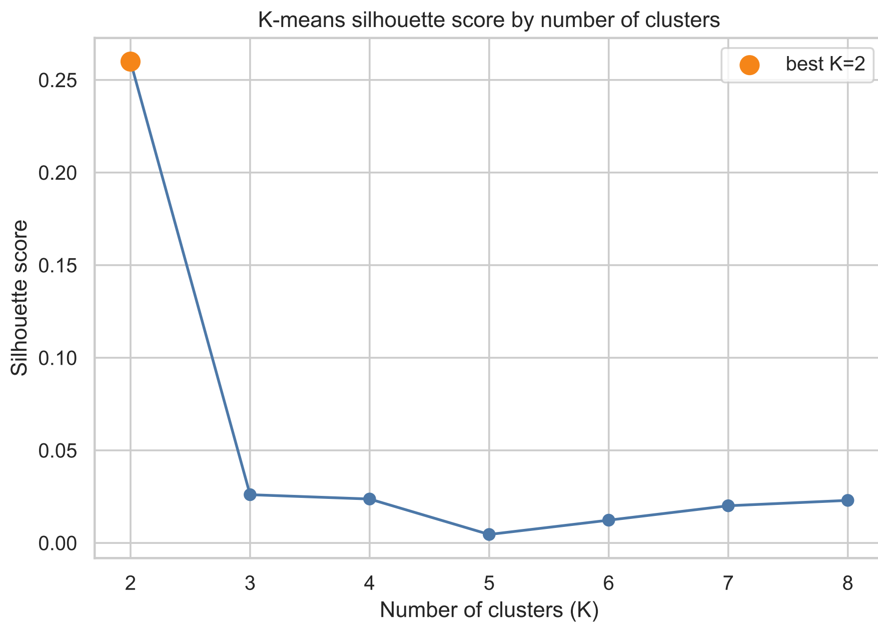

Figure 5: Silhouette score for $K \in \{2, \ldots, 8\}$; the marked maximum selects $K = 2$.

PCA is fitted only to display these clusters in two dimensions [4]. It is not used to transform predictors for logistic regression, the decision tree, or SVC.

4 Main analysis: objective and methods adopted

Logistic regression, an entropy decision tree, and SVC are compared using manual, deterministic, stratified five-fold cross-validation on the raw fit partition [2]. For each model/parameter/fold evaluation, preprocessing and optional standardisation are fitted on `816` subfold training rows and applied unchanged to `204` subfold validation rows. The criterion declared before test evaluation is highest mean fold balanced accuracy; lower fold standard deviation and then the parameter signature break ties. Table 1 shows that this rule preselects the linear SVC. Test metrics do not enter this decision.

Model	CV bal. acc.	CV std.	Selected hyperparameters
SVC	0.6253	0.0437	`C=0.1; balanced; linear`
Logistic regression	0.6176	0.0545	`C=0.1; balanced`
Decision tree	0.5575	0.0691	`alpha=0.01; balanced; leaf=1`

Table 1: Model-family and hyperparameter selection on the fit partition.

The weighted-minus-unweighted CV balanced-accuracy differences are `0.1228` for SVC, `0.0943` for logistic regression, and `0.0385` for the tree. The full comparisons are in `report/tables/`

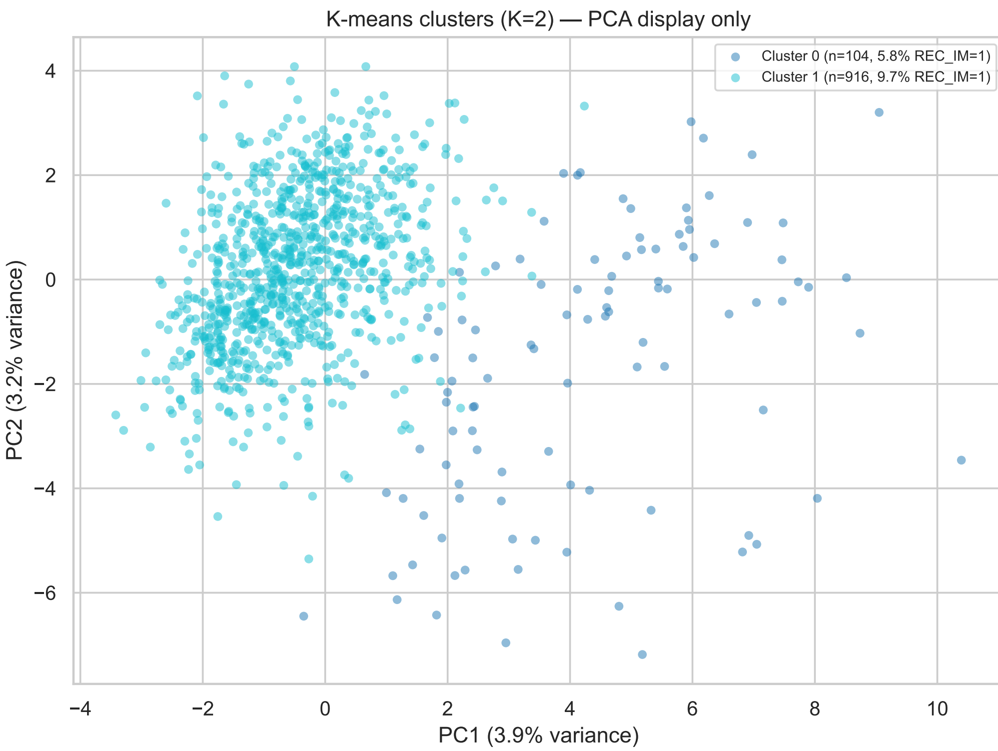

Figure 6: Two-dimensional PCA display of the fitted K-means clusters. PCA is used only for this visualisation and is not part of supervised modeling.

`baseline_vs_weighted.csv`. Tree pruning is shown in Figure 7.

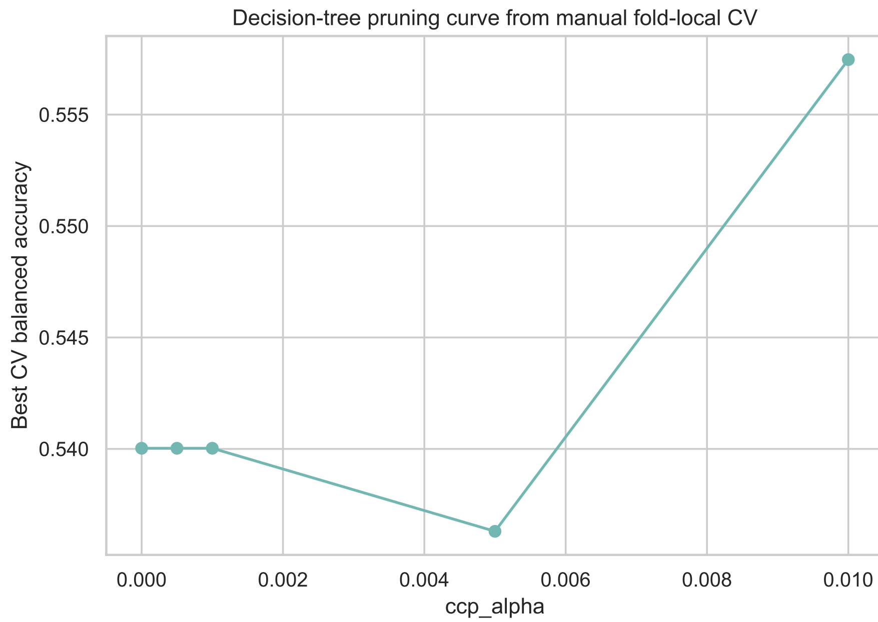

Figure 7: Best decision-tree CV balanced accuracy at each searched `ccp_alpha`.

Only the CV winner is evaluated on validation. Its threshold is selected by balanced accuracy, followed by recall, precision, F1, and proximity to the default threshold. The selected SVC threshold is `0.5056`; validation balanced accuracy is `0.6246`, recall `0.4375`, precision `0.1944`, and F1 `0.2692`. Figure 8 therefore shows exactly one model.

After selection, one preprocessor, scaler, and linear SVC are fitted on the fit partition only. Validation is transformed by these objects to choose the threshold, and no refit follows. Test is then transformed by the same objects and evaluated once. The per-fold evidence is exported in `report/tables/cv_fold_preprocessing_audit.csv`; the outer stage record is exported in `report/tables/evaluation_protocol.csv`.

5 Preview/summary of results

The preselected SVC is the only model evaluated on test. At threshold `0.5056`, it obtains accuracy `0.7471`, balanced accuracy `0.5244`, recall `0.2500`, precision `0.1143`, F1 `0.1569`, and ROC-AUC `0.5597`. The operating point detects 8 of 32 relapse cases and generates 62 false positives. No test-set ranking of model families is performed.

6 Detailed results

Model	Accuracy	Bal. acc.	Recall	Precision	F1	ROC-AUC
SVC	0.7471	0.5244	0.2500	0.1143	0.1569	0.5597

Table 2: The single final evaluation on the isolated test partition.

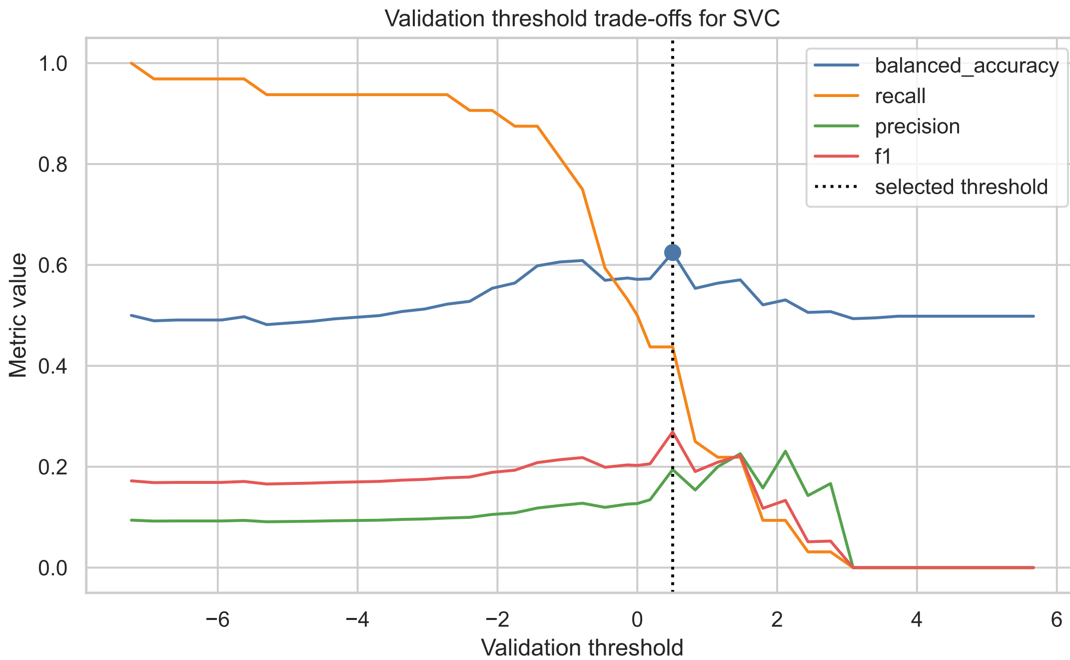

Figure 8: Validation threshold sweep for the CV-preselected SVC. Competing families are not evaluated on validation; the threshold is selected from the same fitted model that is later evaluated on test.

The test confusion counts are TN = 246, FP = 62, FN = 24, and TP = 8. Figure 9 uses the same continuous SVC scores as Table 2; it does not include unselected model families.

Balanced accuracy modestly exceeds chance, while precision remains low because the selected operating point favours relapse sensitivity in an imbalanced sample. The ROC-AUC of `0.5597` similarly indicates weak ranking discrimination. These values are descriptive estimates for the fixed analysis and do not constitute clinical validation.

7 Conclusions

The analysis preserves the admission-time information boundary and completely isolates test data from model-family, hyperparameter, and threshold selection. Every CV fold fits preprocessing and scaling on subfold training rows only. The predeclared criterion selects a linear SVC; one fit-only preprocessor, scaler, and SVC are then reused for validation threshold selection and a single test evaluation without refitting. The resulting balanced accuracy of `0.5244` and ROC-AUC of `0.5597` show that the evaluated admission variables provide only limited signal for `REC_IM`. PCA contributes only a two-dimensional cluster display and does not alter the supervised predictors. Validation is intentionally excluded from final fitting so that its sole supervised role remains threshold selection.

References

[1] S. E. Golovenkin, V. A. Shulman, D. A. Rossiev, P. A. Shesternya, S. Yu. Nikulina, Yu. V. Orlova, and V. F. Voino-Yasenetsky, *Myocardial Infarction Complications*, UCI Machine Learning Repository, 2020. doi: 10.24432/C53P5M.

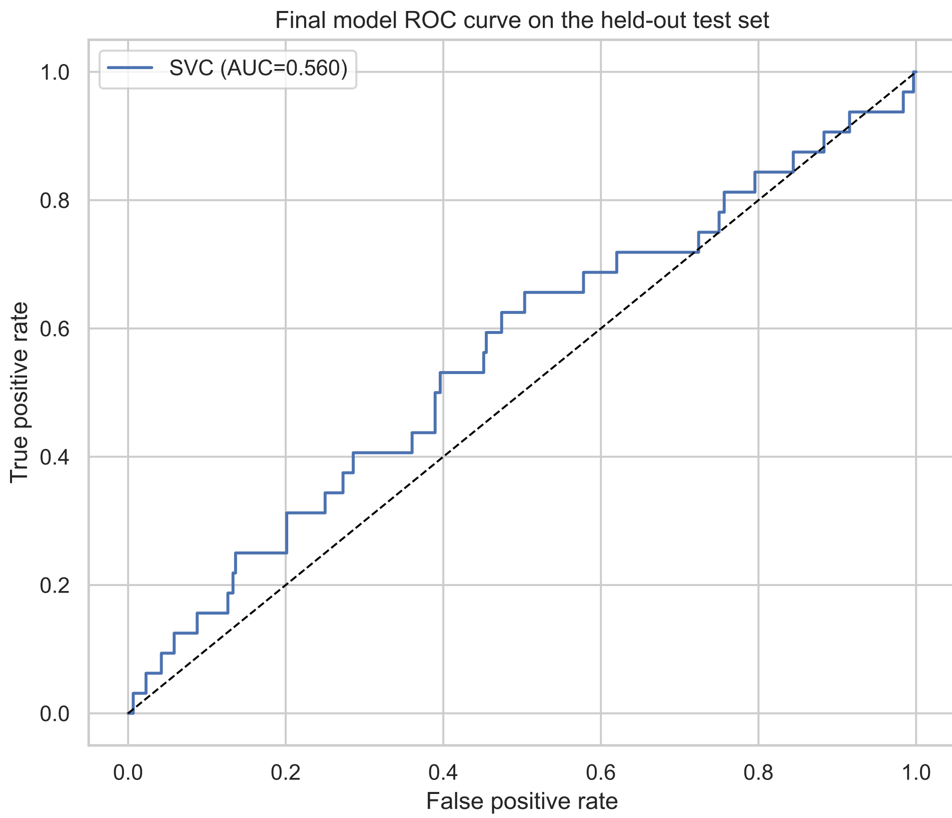

Figure 9: ROC curve for the preselected final SVC on the held-out test set.

[2] A. Manno, *Data Analytics (and Data Driven Decision): Supervised Learning*, Department of Information Engineering, Computer Science and Mathematics, University of L'Aquila, academic year 2025–2026.

[3] A. Manno, *Data Analytics (and Data Driven Decision): Basic Statistics*, Department of Information Engineering, Computer Science and Mathematics, University of L'Aquila, academic year 2025–2026.

[4] A. Manno, *Data Analytics (and Data Driven Decision): Multivariate Statistics and Principal Component Analysis*, Department of Information Engineering, Computer Science and Mathematics, University of L'Aquila, academic year 2025–2026.